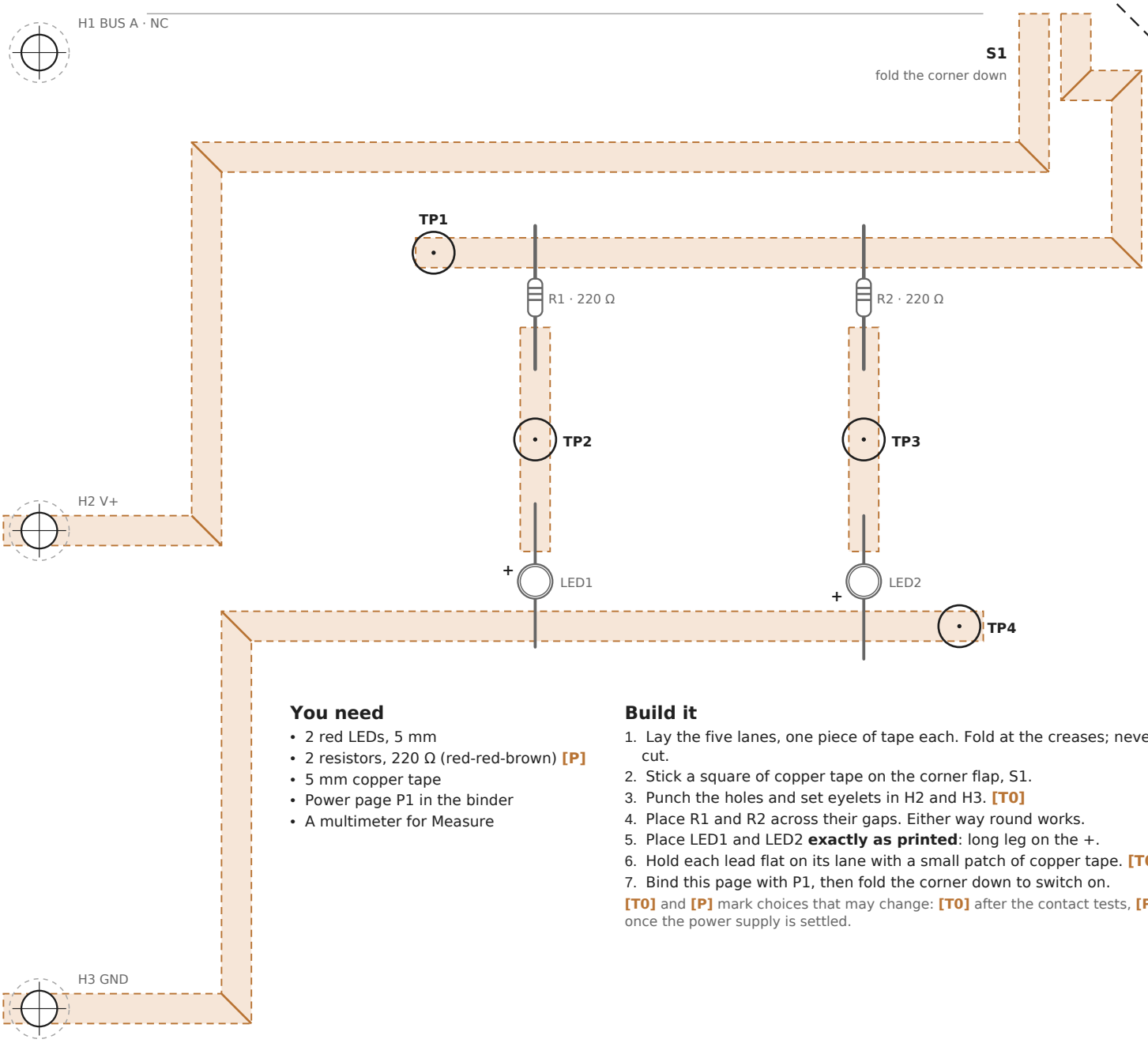
L01 LEDs only work one way

L01

Close the loop, then find out why an LED cares which way round it goes.

Draft: tagged steps may change



You need

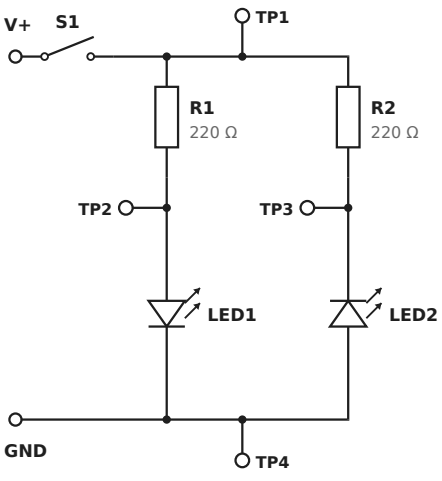
- 2 red LEDs, 5 mm
- 2 resistors, 220 Ω (red-red-brown) **[P]**
- 5 mm copper tape
- Power page P1 in the binder
- A multimeter for Measure

Build it

1. Lay the five lanes, one piece of tape each. Fold at the creases; never cut.
2. Stick a square of copper tape on the corner flap, S1.
3. Punch the holes and set eyelets in H2 and H3. **[T0]**
4. Place R1 and R2 across their gaps. Either way round works.
5. Place LED1 and LED2 **exactly as printed**: long leg on the +.
6. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
7. Bind this page with P1, then fold the corner down to switch on.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Schematic



V+ and GND come from P1 through the binder.

H4 BUS B · NC

Predict LED2 is printed the other way round.

When you fold the corner down, which LEDs will light? Circle one for each.

LED1 lights / stays dark **LED2** lights / stays dark

Why do you think so?

Measure Hold the corner down. Black probe on TP4, red on each point.

TP1 _____ V **TP2** _____ V **TP3** _____ V

Lit? **LED1** Y / N **LED2** Y / N

Explain Why does only one LED light?

What would you change to make the other one light?

Print at 100 %: this bar should measure 100 mm.